THIRD IRIS

AUTONOMY PROPOSAL · 2026

SEES.
TRACKS.
NEEDS NO GPS.

An aerial system that detects, follows and intercepts a moving target — and navigates itself — using nothing but a camera, even under full GPS jamming.

SIM-PROVEN

REAL-DATA TRAINED

HARDWARE-READY PATH

THE PROBLEM

GPS is the single point of failure modern conflict exploits.

GPS-DEPENDENT DRONE

Jammed → blind. Spoofed → hijacked. In contested airspace, satellite navigation is the first thing to die — and the aircraft dies with it.

✕ MISSION LOST

VISION-DRIVEN DRONE

Cameras are passive. Nothing to jam, nothing to spoof, no emissions to detect. If the aircraft can see, it can navigate, track and finish the mission.

✓ MISSION CONTINUES

Ukraine-era electronic warfare made this concrete: drones that rely on GPS are being neutralized by jammers costing less than the airframe. The eye is the backup that cannot be jammed.

SYSTEM CONCEPT

One camera. Two minds.

Every frame is consumed twice — **tracking** (detector → tracker → Kalman → body-frame servo) and **self-localization** (VIO fuses camera + IMU). No absolute position is ever needed, so jamming has nothing to attack.

01 · SEE	02 · TRACK	03 · LOCATE	04 · AVOID	05 · ACT
Detect	**Lock & predict**	**Self-position**	**Dodge**	**Follow / intercept**
YOLO detector + instance segmentation, per frame	ByteTrack + re-ID + image-space Kalman; coasts dropouts	Visual-inertial odometry (VIO / VINS) — zero GPS	Depth-as-LiDAR repulsion — pursuit unbroken	Body-frame visual servoing · RL pilot @ 10 Hz

10 Hz
ONBOARD CONTROL LOOP

2
INDEPENDENT PIPELINES, ONE SENSOR

0
SATELLITES REQUIRED

0.80 / .97
DETECT MAP · REAL / SIM

PERCEPTION

Trained on the real threat.

Not just simulation. The detector is fine-tuned on ~50,000 real images of Shahed-type loitering munitions and quadcopters — plus a bird class so it never chases the wrong wings.

0.77

MAP50 · REAL HELD-OUT

0.97

MAP50 · SIM TARGET

DATASETS · SHAHED 50K / MJOLNIR 1.2K / AIRSIM SYNTH

CLASSES · DRONE + BIRD (FALSE-LOCK FILTER)

It copies a master. Then it beats him.

- Segment & filter — real + synthetic footage is mask-labeled; birds, clutter and shadows are explicitly trained out.
- Imitate — a neural pilot learns by copying a hand-built expert across thousands of randomized pursuit episodes (the Skydio recipe).
- Exceed — reinforcement learning then trains past the teacher under realistic flight dynamics, noise and detector drop-outs.
- Prove — every build re-flies orbit, zigzag, weave, climb and dive scenarios in simulation. Numbers or it didn't happen.

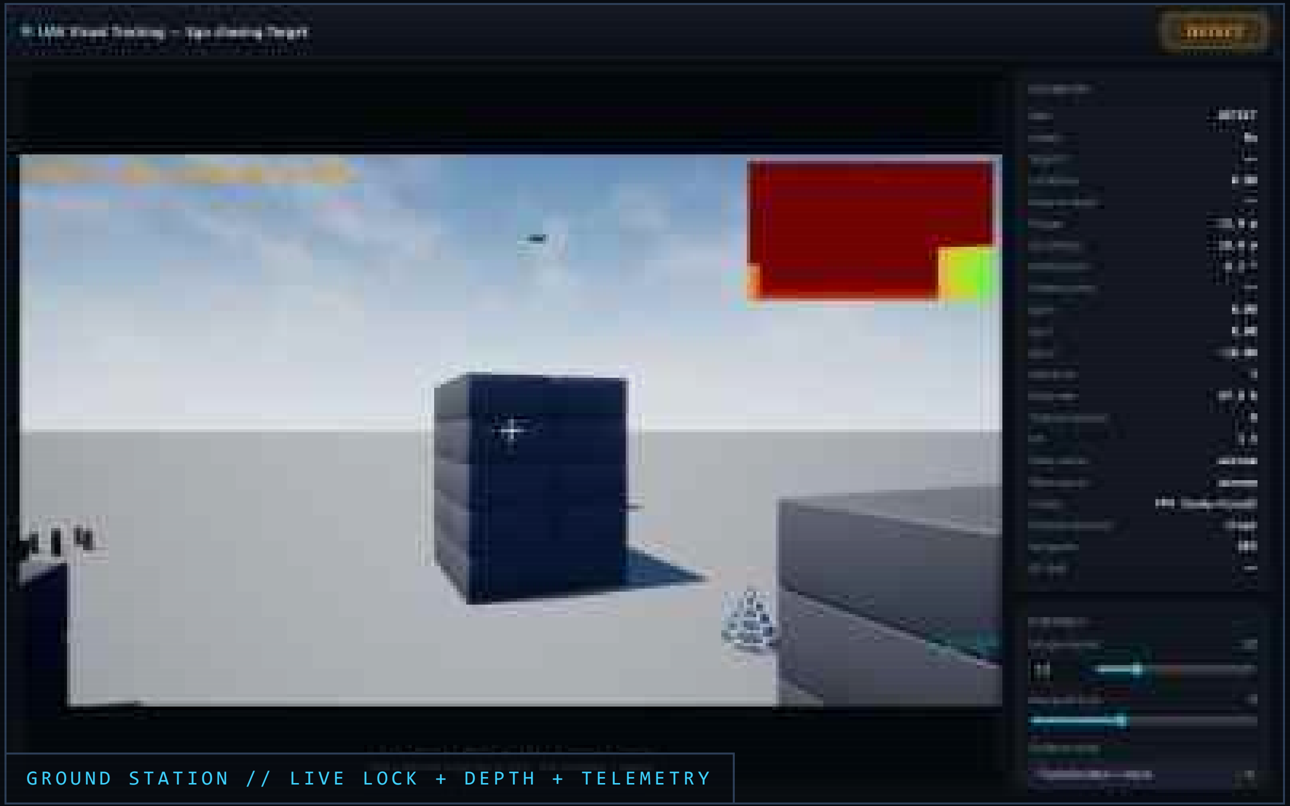
PIPELINE · DATA → SEGMENT → FILTER → IMITATE → RL → SIM-TEST

METRIC	HAND-BUILT EXPERT	LEARNED PILOT
Target kept in frame	100%	99.9%
Centering error	0.263	0.107 · 2.5× tighter
Stand-off hold error	7.2 m	4.3 m · 40% better
Survives detector drop-outs	partially	yes — trained on them

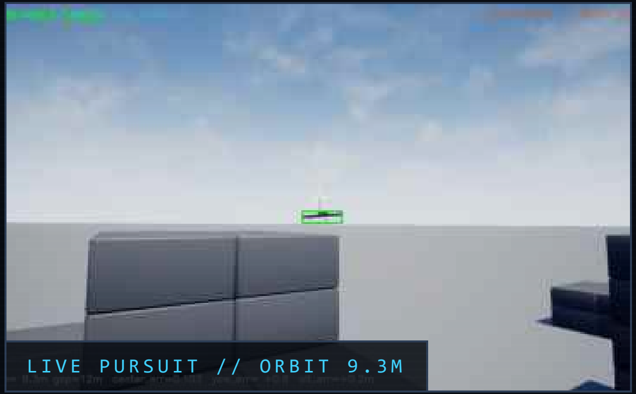
300 IDENTICAL EPISODES · REALISTIC ACTUATOR LAG · MEASURED, NOT CLAIMED

Click the target. The aircraft does the rest.

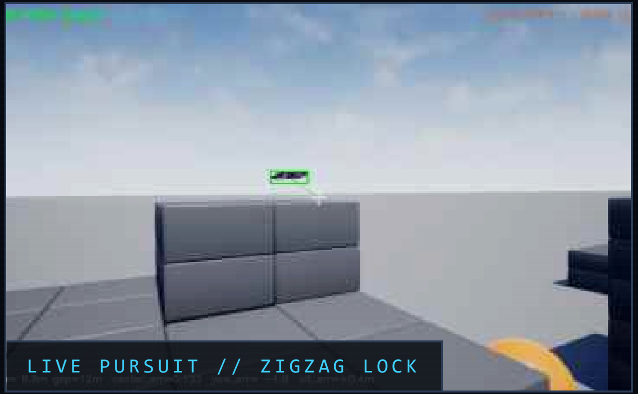
- Click-to-lock in a live ground-station web UI — one operator, one click.
- Predictive lock — a motion model leads the target and coasts straight through detection gaps.
- Stand-off hold — keeps a chosen distance; speeds match, no spiral, no overshoot.
- ~90% in-frame through aggressive evasive maneuvers, vision only.



GROUND STATION // LIVE LOCK + DEPTH + TELEMETRY



LIVE PURSUIT // ORBIT 9.3M



LIVE PURSUIT // ZIGZAG LOCK

LIVE RUN // PURSUIT PATH + ALT HOLD

SELF-LOCALIZATION

It knows where it is. Without being told.

- Visual-inertial odometry — camera motion + IMU fused into a live position estimate; the same math inside VINS-class flight systems.
- Depth sensing — true metric distance to the scene and the target.
- SLAM-style drift correction — recognizing seen places snaps the estimate back.
- Compass-bounded heading — yaw can't silently wander.

GPS ▸ ON

VISION + SATELLITES FUSED

GPS ▸ JAMMED

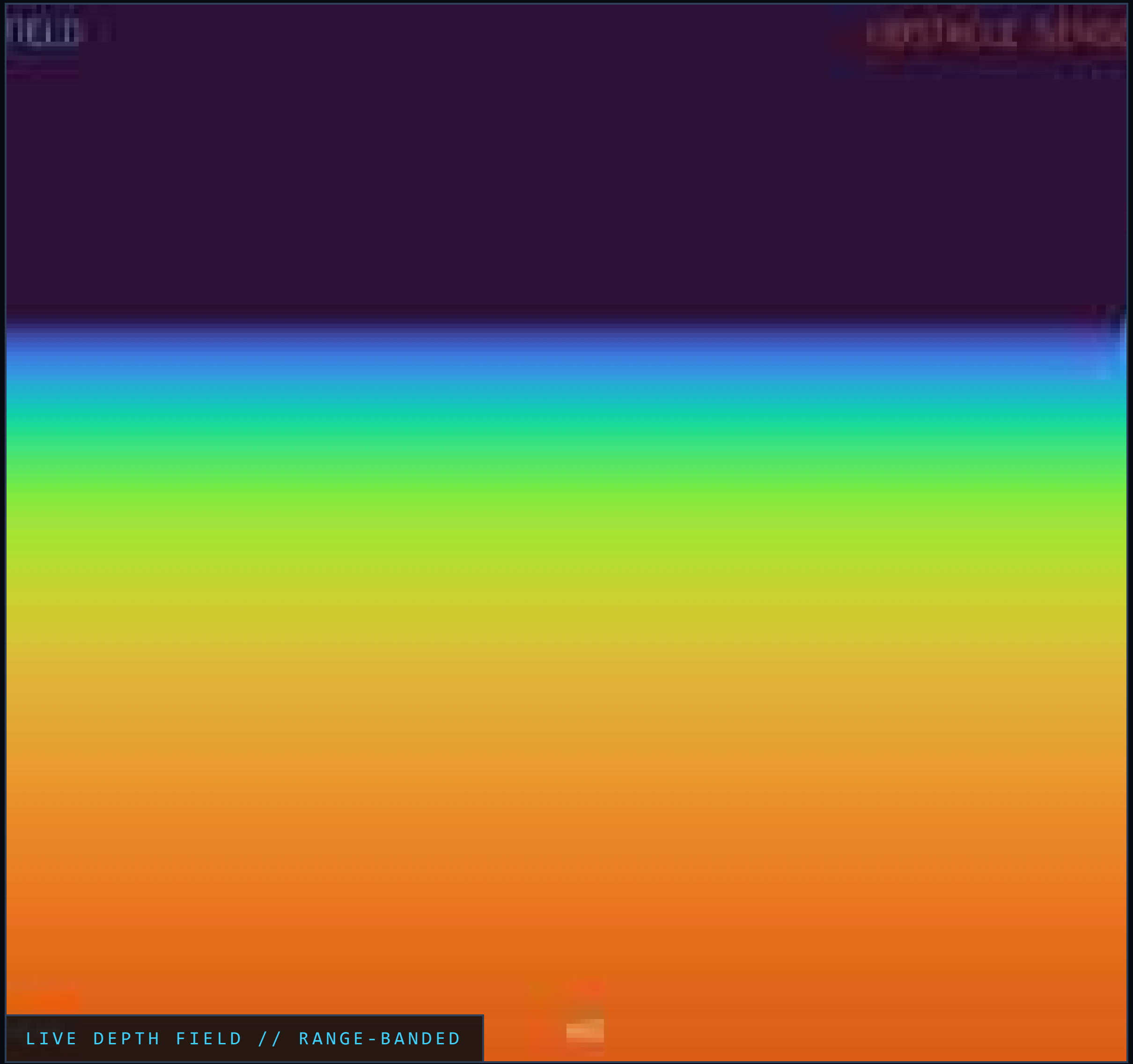
SEAMLESS SWITCH TO VISION-ONLY

FLIGHT SAFETY

Dodges the world. Keeps the lock.

- Depth-as-LiDAR — the depth image is read as a wall of distances, hundreds of rays per frame, no extra sensor.
- Reactive steering — repulsion from close obstacles bends the flight path around buildings, terrain, wires.
- Pursuit-preserving — avoidance and tracking blend; the target stays framed while the aircraft threads the gap.
- Zero collisions across the full simulated scenario suite.

RUNS FULLY ONBOARD · NO PRIOR MAP · NO RF EMISSIONS



LIVE DEPTH FIELD // RANGE-BANDED

APPLICATIONS

Two markets. Same eye.

COMMERCIAL — DUAL-USE / CIVIL

- Follow-me cinematography — locked, smooth subject tracking with no pilot skill.
- GPS-poor inspection — under bridges, inside plants, beside steel structures where GPS multipaths.
- Search & rescue — canyons, forests, indoors; tracks a moving person.
- Warehouse autonomy — indoor flight with zero satellite coverage by design.
- Urban delivery — canyon-tolerant navigation between towers.

DEFENSE — CONTESTED EW ENVIRONMENT

- Counter-UAS interception — vision-locked pursuit of Shahed-class and quadcopter threats.
- ISR under jamming — surveillance continues when every satellite channel is denied.
- Convoy & perimeter overwatch — autonomous escort with click-to-task.
- Terminal guidance — proportional-navigation strike profile, vision-only end-game.
- Comms-denied missions — decides onboard; no link, no leash.

A System-2 mind. On the aircraft.

A small on-board language model supervises the mission — it **advises**, it never flies. The 10 Hz pilot, tracker and VIO stay untouched.

SLOW SUPERVISOR · ~0.5 HZ · ADVISORY

Qwen2.5-1.5B · 4-bit · Jetson Orin

Reads telemetry text (track status, VIO health, GPS-jam flag, battery, geofence) → emits an enum mode + JSON. ~1.5 s / turn.

`{ mode: SEARCH | TRACK | STRIKE | RETURN | HOLD }`

▼

DETERMINISTIC SAFETY GATE · FSM

Validates the proposal, allows only legal mode transitions, clamps to the safety envelope (geofence / altitude / battery-RTL / speed). Stale or invalid → safe default. The agent cannot command actuators.

▼

FAST STACK · 10 HZ · UNCHANGED

YOLO + ByteTrack → BC+RL pilot → PX4 / MPC. VIO → state estimate. The pilot IS System-1 — instant, sub-millisecond.

WHY NOT A VLA

- 7B VLAs (OpenVLA, π0) replace the pilot and don't run real-time on Orin.
- We keep our fast policy and add only the System-2 idea on top.
- Text-over-telemetry, not raw pixels — far less hallucination.

GROUNDED IN

Hi Robot — Physical Intelligence, 2025
LLM-Land — on-device LLM + MPC, 2025
NVIDIA GR00T — System-2 / System-1 split

FINAL PHASE · BUILDS IN SIM LAST

STATUS & ROADMAP

Same software, sim to sky.

DONE · SIMULATION

Full stack proven in AirSim

Detect · track · follow · avoid · GPS-jam handover · strike profile — all measured across maneuver suites.

DONE · REAL DATA

Detector + learned pilot trained

50k real Shahed/quad images; RL pilot beats the hand-built expert 2.5× on centering.

NEXT · PRODUCTION VIO

OpenVINS → flight controller

Production-grade odometry fused into PX4; sub-meter drift target on benchmark flights.

THEN · HARDWARE

RealSense + Jetson Orin

Identical code, real sensor. Depth camera + edge GPU + gimbal on a flight-test airframe.

If it can see, it can fight.

Vision-only autonomy — jam-proof by physics, not by promise.

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